Kristopher Jung

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience building distributed systems, high-throughput event pipelines, and data platforms at scale across Capital One, Criteo, and General Motors. Proven ability to own complex backend systems end-to-end, from architecture through production operations, with recent hands-on work in AI-powered observability and ETL at the enterprise level.

EXPERIENCE

Senior Software Engineer (Principal Associate) Apr 2025 – Present
Capital One *Plano, TX*

- Architect and implement distributed data platforms and ETL pipelines supporting credit decisioning and risk prediction across Capital One's small-business lending systems using Airflow, PySpark, Databricks, Snowflake, and AWS.
- Build high-performance APIs with Python and FastAPI, integrating Finicity to enable open banking and seamless external bank account connections for millions of Capital One customers.
- Build a RAG pipeline using a vector store with LlamaIndex, powering an AI-driven data observability platform that automated anomaly detection and root-cause analysis, reducing MTTR by 60%, reclaiming 1,800+ engineering hours, and lowering operational costs by $200K+ annually.
- Automate engineering workflow reporting by integrating Claude Skills with Jira, Confluence, and Jenkins, improving development visibility and reducing manual status coordination across cross-functional teams.

Senior Software Engineer (Promoted from Software Engineer II) May 2022 – Apr 2025
Criteo *Ann Arbor, MI*

- Scaled a distributed ad-campaign budgeting and billing platform to 53K+ events/sec and $1.5B+ in transactions, supporting 17,000+ advertisers and 200+ retailers.
- Designed and implemented Kafka-based event pipelines, C#/.NET microservices, and T-SQL data stores for a high-throughput Ledger system.
- Led end-to-end credit-card payments integration, enabling self-serve ad campaigns for small advertisers and reducing settlement latency from 30 days to near real-time.
- Built Grafana dashboards and Kibana alerting to detect transaction anomalies across 25,000+ active campaigns, reducing time-to-mitigation during incidents.
- Led production incident response for billing and payment services, coordinating diagnosis and mitigation to protect revenue-critical transaction workflows.
- Mentored interns and early-career engineers through code reviews, design discussions, and pair programming.

Software Engineer Feb 2020 – May 2022
General Motors *Detroit, MI*

- Developed Java/Spring microservices and Kafka/Pulsar event pipelines supporting vehicle ordering and inventory systems for GM's network of 4,500+ dealerships across the U.S., Canada, and Mexico.
- Built a Redis-backed rule engine for vehicle pattern management, improving decision consistency and system responsiveness.
- Implemented CI/CD pipelines using Azure DevOps, automating build, test, and deployment workflows and reducing release errors.
- Built TypeScript/Angular UI components enabling order and pattern management workflows integrated with backend services.

SKILLS

Languages: Python, Java, SQL, C#, TypeScript

Backend & Distributed Systems: FastAPI, Spring Boot, .NET, Kafka, Pulsar, Redis, Airflow, PySpark

Data & AI: Databricks, Snowflake, LlamaIndex, RAG

Cloud & Platforms: AWS, DynamoDB, S3, MongoDB

DevOps & Observability: Docker, Jenkins, Grafana, Kibana, CI/CD

EDUCATION

Indiana University Bloomington, IN
Bachelor of Science in Computer Science. *Dec 2019*